

Introduction to Numerical Weather Prediction



Chapter 5: Numerical Weather Prediction Models



Objective

- *To Differentiate Grid point and Spectral models*
- *To Identify the Hydrostatic and non-hydrostatic models*
- *To Differentiate barotropic and baroclinic model*
- *To Describe Primitive Equation (PE) model*
- *To Identify the Sources of Errors in NWP*

Numerical Weather Prediction Models

Model Types

Introduction

- ❑ The "model type" refers to whether a model represents meteorological data as point-like (gridpoint) or as a series of waves (spectral), and hydrostatic or non hydrostatic.
- ❑ The type of model being used (grid point or spectral) influences:
 1. How the model equations are solved
 2. How the data are represented
 3. The type of weather features that can be resolved

MODEL TYPES

- ☐ *Grid point and spectral models*
- ☐ *Hydrostatic and non-hydrostatic models*
- ☐ *Equivalent barotropic model and baroclinic model*
- ☐ *Primitive Equation (PE) model*

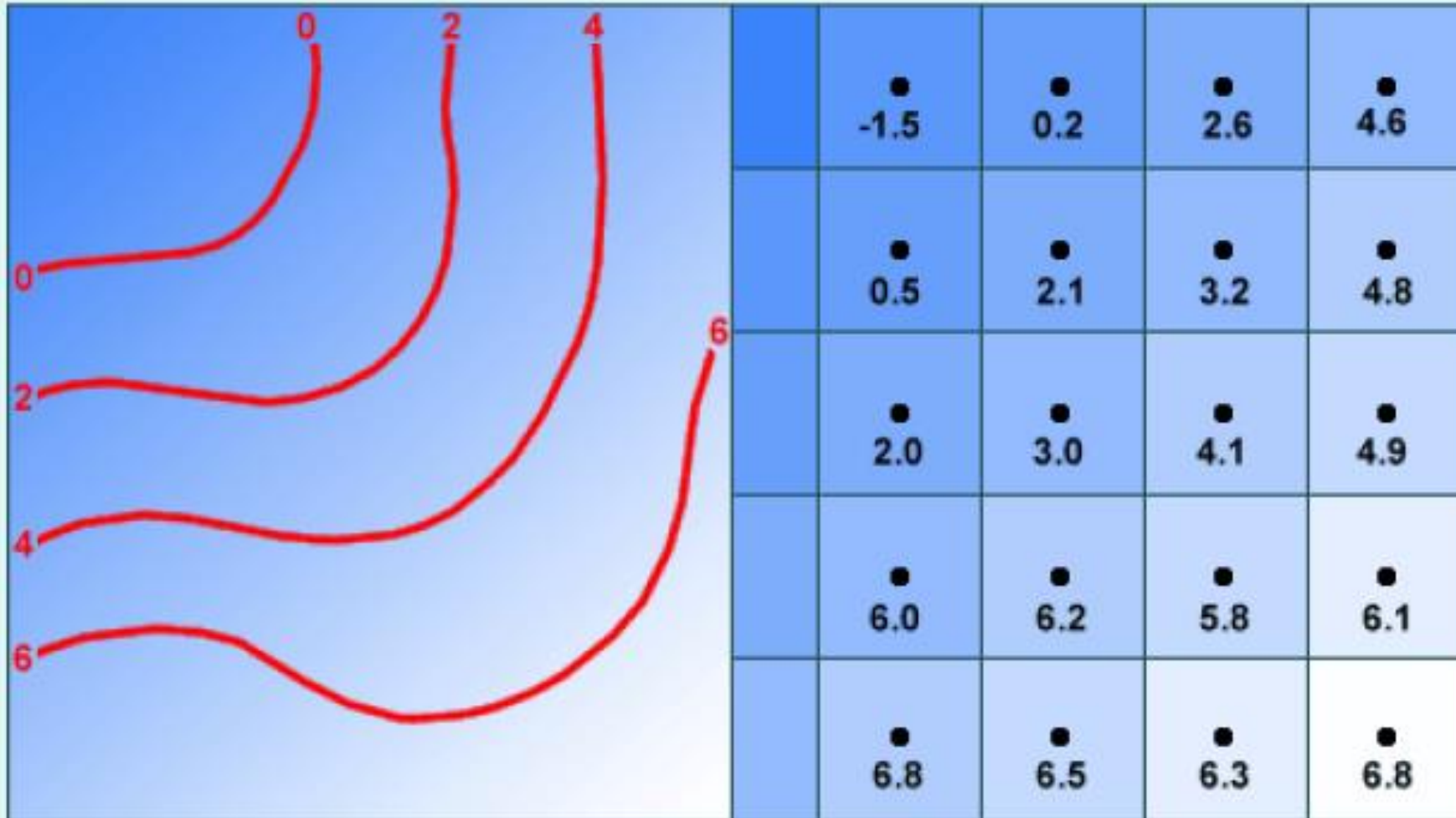
Grid point model

GRID POINT MODELS

- The arrangement of variables within and around the grid cube (called a staggered grid) has advantages when calculating derivatives.
- Average Thermodynamic-properties inside the grid cube are represented at the center, while the winds on the faces are associated with fluxes into and out of the cube.

Data Representation

- ❖ In the real atmosphere, wind, pressure, temperature, and moisture vary from location to location in a smooth, continuous way.
- ❖ Grid point models, however, perform their calculations on a fixed array of spatially disconnected grid points. The values at the grid points actually represent an area average over a grid box.
- ❖ The continuous temperature field shown in the next graphic, therefore, must be represented at each grid point as shown by the black numbers in the right panel of the previous graphic. The temperature value at the grid point represents the grid box volume average.



Actual smooth and continuous temperature field in degrees C (similar to spectral model representation)

Grid point model representation of the same temperature field in degrees C

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Truncation Errors

- ❑ Grid point models must use finite difference techniques to solve the forecast equations.
- ❑ In the real atmosphere, advection often occurs at very small scales.
- ❑ The greater the distance between grid points, the less likely the model will be able to detect small-scale variations in the temperature and moisture fields.
- ❑ The lack of resolution introduces errors into the solution of the finite difference equation.

❑ Deficiencies in the ability of the finite difference approximations to calculate gradients and higher order derivatives exactly are called **TRUNCATION ERRORS.**

Characteristics

- ❑ Data are represented on a fixed set of grid points.
- ❑ Resolution is a function of the grid point spacing.
- ❑ All calculations are performed at grid points.
- ❑ Finite difference approximations are used for solving the derivatives of the model's equations.
- ❑ E.g WRF

- ❑ Truncation error is introduced through finite difference approximations of the primitive equations.
- ❑ The degree of truncation error is a function of grid spacing and time-step interval.

Advantages

- Can provide high horizontal resolution for regional and meso-scale applications.
- Do not need to transform physics calculations to and from gridded space.
- As the physics in operational models becomes more complex, grid point models are becoming computationally competitive with spectral models.
- Non-hydrostatic versions can explicitly forecast details of convection, given sufficient resolution and detail in the initial conditions.

Dis-advantages

- Finite difference approximations of model equations introduce a significant amount of truncation error.
- Small-scale noise accumulates when equations are integrated for long periods.
- The magnitude of computational errors is generally more than in spectral models of comparable resolution.
- Boundary condition errors can propagate into regional models and affect forecast skill.
- Non-hydrostatic versions cover only very small domains and short forecast periods.

Spectral Models

Data Representation

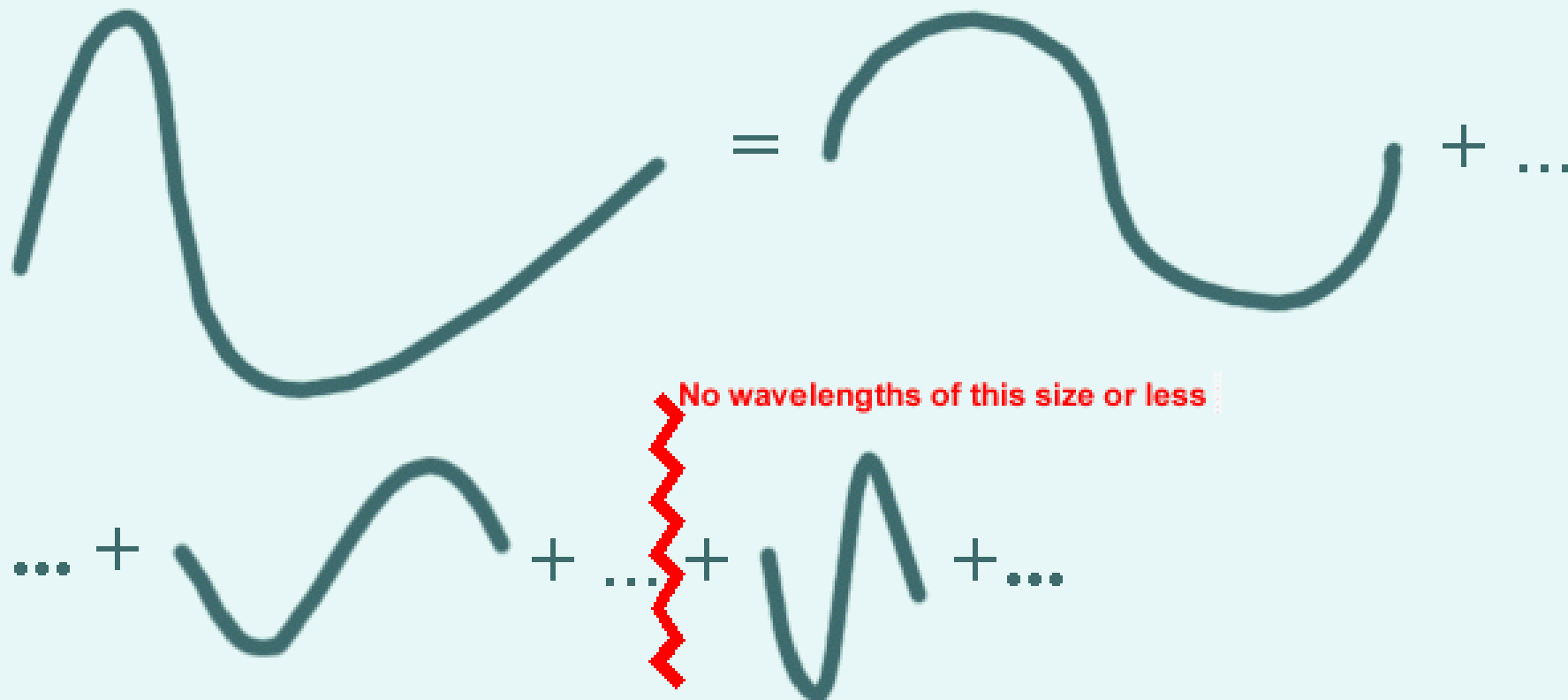
❑ Spectral models represent the spatial variations of meteorological variables as a finite series of waves of differing wavelengths.

Truncation Errors

What are the effects of truncation in a spectral model?

- ❑ Recall that in a grid point model, truncation error is associated with the finite difference approximations used to evaluate the derivatives of the model forecast equations.
- ❑ One of the nice features of the spectral formulation is that most horizontal derivatives are calculated directly from the waves and are therefore extremely accurate.
- ❑ This does not mean that spectral models have no truncation effects at all. The degree of truncation for a given spectral model is associated with the *scale of the smallest wave represented by the model.*

Truncation Effects In Spectral Models



Any wavelengths equal to or less than the wave to the right of the red jagged line will not be resolved by the spectral model. This results in what is known as truncation effects.

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❑ A grid point model tries to include all scales but does a poor job of handling waves only a few grid points across. A spectral model represents all of the waves that it resolves perfectly but includes no information on smaller-scale waves.

❑ *If the number of waves in the model is small (for example, T80), only larger features can be represented and smaller-scale features observed in the atmosphere will be entirely eliminated from the forecast model.*

❑ *Therefore, spectral models with limited numbers of waves can quickly depart from reality in situations involving rapid growth of initially small-scale features.*

❑ Several types of wave orientation are possible in spectral models. Triangular (T, as in T170) configuration is the most common in operational models since it has roughly the same resolution in the zonal and meridional directions around the globe.

Characteristics

- ❑ Data are represented by wave functions.
- ❑ Resolution is a function of the number of waves used in the model.
- ❑ Model resolution is limited by the maximum number of waves.
- ❑ The linear quantities of the equations of motion can be calculated without introducing computational error.
- ❑ E.g GFS, ECMWF

- ❑ Grids are used to perform non-linear and physical calculations.
- ❑ Transformations occur between spectral and grid point space.
- ❑ Equations can be integrated for large time steps and long periods of time.
- ❑ Originally designed for global domains

Advantages

- ❑ The magnitude of computational errors in dynamics calculations is generally less than in grid point models of comparable resolution.
- ❑ Can calculate the linear quantities of the equations of motion exactly.
- ❑ At horizontal resolutions typically required for global models (late 1990s), require less computing resources than grid point models with equivalent horizontal resolution and physical processes

Disadvantages

- ❑ Transformations between spectral and grid point physics calculations introduce errors in the model solution.
- ❑ Generally not designed for higher resolution regional and mesoscale applications.
- ❑ Computational savings decrease as the physical realism of the model increases

Hydrostatic Models

- ❑ Most grid point models and all spectral models in the current operational NWP models are hydrostatic. This means that no vertical accelerations are calculated explicitly.
- ❑ The hydrostatic assumption is valid for synoptic-and planetary-scale systems and for some mesoscale phenomena. An important exception is deep convection, where buoyancy becomes an important force.
- ❑ Hydrostatic models account for the effects of convection using statistical parameterizations approximating the larger-scale changes in temperature and moisture caused by non-hydrostatic processes.

Characteristics

- Use the hydrostatic primitive equations, diagnosing vertical motion from predicted horizontal motions.
- Used for forecasting synoptic-scale phenomena, can forecast some mesoscale phenomena.
- Used in both spectral and grid point models
- E.g All global and regional scale models

Advantages

- ❑ Can run fast over limited-area domains, providing forecasts in time for operational use.
- ❑ The hydrostatic assumption is valid for many synoptic-and sub-synoptic-scale phenomena

Disadvantages

- ❑ Cannot predict vertical accelerations
- ❑ Cannot predict details of small-scale processes associated with buoyancy

Non-Hydrostatic Models

- ❑ Currently, most non-hydrostatic models are grid point models. They are generally used in forecast or research problems requiring very high horizontal resolution (from tens of meters to a few kilometers) and cover relatively small domains.
- ❑ Non-hydrostatic models can explicitly (directly) forecast the release of buoyancy in the atmosphere and its effects on the development of deep convection.
- ❑ To do this, non-hydrostatic models must include an additional forecast equation that accounts for vertical accelerations and vertical motions directly, rather than determining the vertical motion diagnostically from horizontal divergence.

- ❑ To account for vertical motions and buoyancy properly, non-hydrostatic models must include a great deal of detail about cloud and precipitation processes in their temperature and moisture forecast equations.
- ❑ Since hydrostatic models do not have a vertical motion forecast equation, none of these processes can directly affect the vertical motion in their predictions.
- ❑ One disadvantage of non-hydrostatic models is *longer computation time*. Since the models must finish running in time for forecasters to use model products, hydrostatic models are more advantageous unless non-hydrostatic phenomena need to be simulated or unless resolution finer than around 10 km is needed.
- ❑ Non-hydrostatic models run at very high resolution characteristically predict detailed mesoscale structure and associated forecast impacts on surrounding areas.

Characteristics

- ❑ Use the non-hydrostatic primitive equations, directly forecasting vertical motion.
- ❑ Used for forecasting small-scale phenomena.
- ❑ Predict realistic-looking, detailed mesoscale structure and consistent impact on surrounding weather, resulting in either superior local forecasts or large errors

Advantages

- ❑ Calculate vertical motion explicitly
- ❑ Explicitly predict release of buoyancy
- ❑ Account for cloud and precipitation processes and their contribution to vertical motions
- ❑ Capable of predicting convection and mountain waves

Disadvantages

- ❑ Take longer to run than hydrostatic models with the same resolution and domain size.
- ❑ Used for limited-area applications, so they require boundary conditions (BCs) from another model; if the BC slack the structure and resolution characteristic of fields developing inside the model domain, they may exert great influence on the forecast.
- ❑ May predict realistic-looking phenomena, but the timing and placement may be unreliable

Barotropic Model:

- In normal condition Isobars (Equal pressure lines) are parallel to the Isotherms (Equal temperature lines) in the atmosphere.
- This condition known as **Barotropic** condition.
- The Barotropic condition will exists at anti-cyclonic condition and calm winds and normal pressure levels.

Barotropic and Baroclinic Model

- Barotropic atmosphere (constant density/temperature on pressure surface, no vertical motion)
- Absolute vorticity conserved

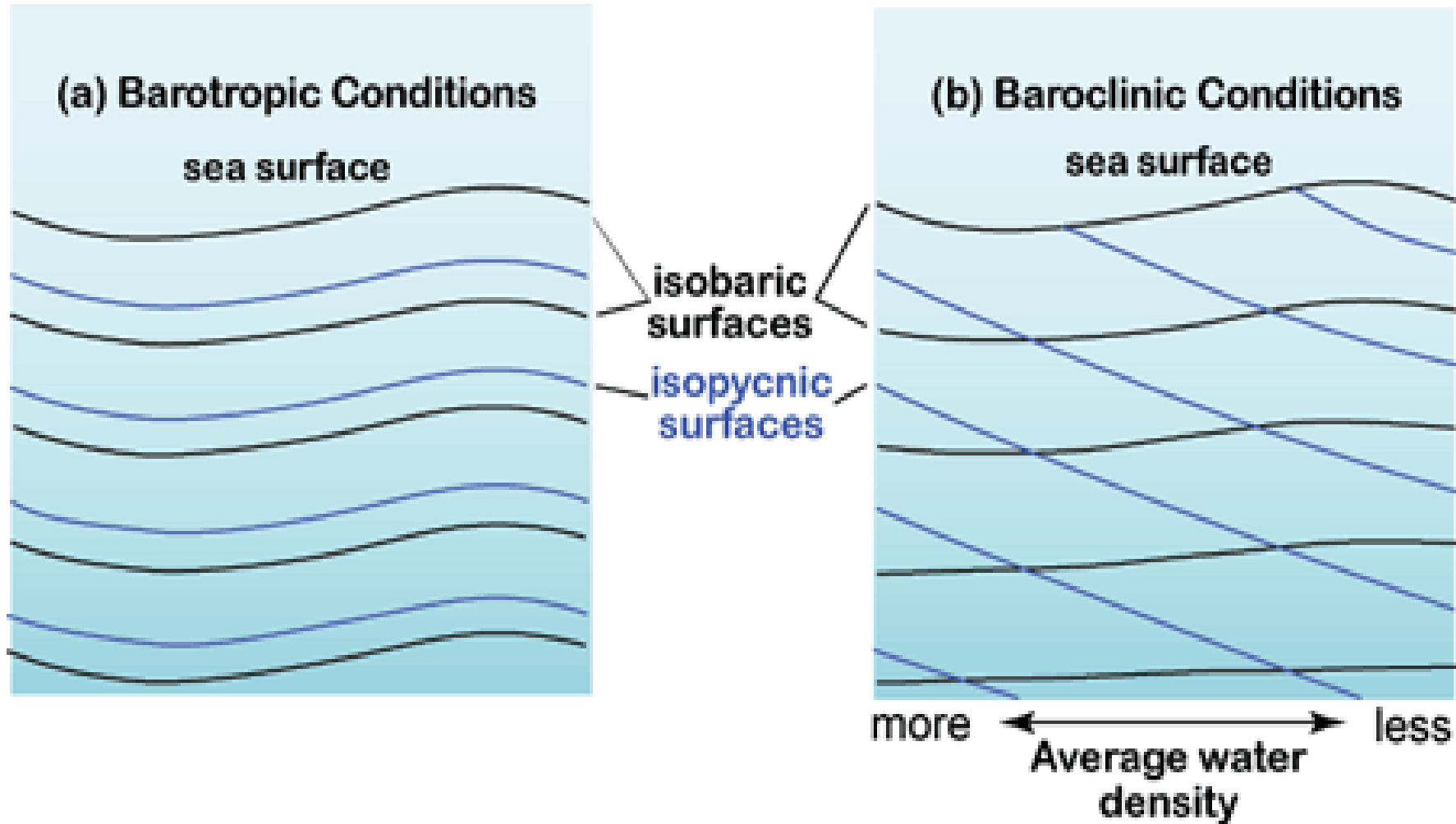
$$\frac{d(f + \zeta)}{dt} = 0$$

- Somewhat skillful at large-scale wave prediction

Baroclinic Model

- At abnormal conditions with cyclonic flow and pressure gradient conditions, the isobars are inclined (intercepting), this condition in the atmosphere known as **Baro-clinic** condition.

Barotropic and Baroclinic



Primitive Equation Models

- This model is generally considered better than the Barotropic model in the sense that it allows for divergence to occur at its free surface.
- It is a single level model and is based on the primitive equations.
- The model is also known as the 'Barotropic Divergent Model' or the 'Shallow Water Equations Model'.

Primitive Equation Models

- The “primitive equations” are fundamental relationships that govern atmospheric motion
 - ✓ Equations of motion (3) (cons. of momentum)
 - ✓ Continuity equation (cons. of mass)
 - ✓ First law of thermodynamics (cons. of energy)
 - ✓ Moisture equation (cons. of moisture)
 - ✓ Equation of state

How Can Forecasts Go Bad?

The Prediction Phase – How Can Forecasts Go Bad?

- There are 2 main sources of error in NWP forecasts:

- 1. Initial condition error**

- 2. Physics/model errors**

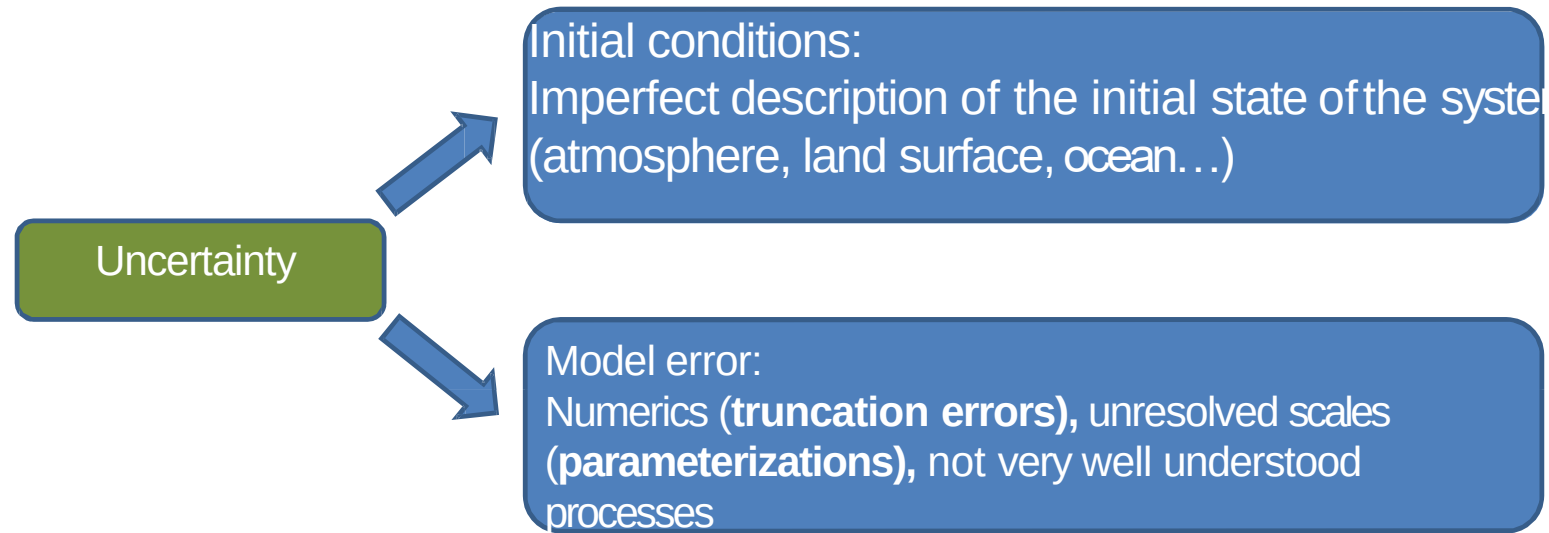
Sources of error in NWP

- Errors in the initial conditions
- Errors in the model
- Intrinsic predictability limitations

Sources of Errors

<u>Initial Condition Errors</u>	<u>Model Errors</u>
1 Observational Data Coverage - a Spatial Density - b Temporal Frequency	1 Equations of Motion Incomplete
2 Errors in the Data - a Instrument Errors - b Representativeness Errors	2 Errors in Numerical Approximations - a Horizontal Resolution - b Vertical Resolution - c Time Integration Procedure
3 Errors in Quality Control	3 Boundary Conditions - a Horizontal - b Vertical
4 Errors in Objective Analysis	4 Terrain/Topography
5 Errors in Data Assimilation	5 Physical Processes
6 Missing Variables	

Forecast uncertainty



Both are important sources of errors in the forecast

Error growth due to the chaotic nature of the atmosphere amplifies both sources of error.

Good Numerical Forecasts Require...

- Initial conditions that adequately represent the state of the atmosphere (three-dimensional wind, temperature, pressure, moisture and cloud parameters)
- Numerical weather prediction model that adequately represents the physical laws of the atmosphere over the whole globe.

Summary: NWP

- To produce an accurate forecast.

Steps:

- Collecting Data
- Quality Control of Data
- Objective Analysis & Grid Initialization
- Numerical Modeling
- Output Grids, Statistical Guidance
- Human Forecast

Summary: Numerical Weather Modeling

- **Dynamical Equations:**
- **Numeric:** Solve prognostic equations using finite differencing (grid point models) or actual derivatives (spectral)
- **Parameterizations** needed to represent processes not governed by dynamics:
 - ✓ Subgrid scale processes
 - ✓ Radiation/convection
 - ✓ Land surface interactions

....END

Thanks !